

Confining iodide migration with quantified barrier for durable perovskite solar cells

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The migration of iodide ions out of the perovskite film degrades charge transport and electrode layers, reducing the long-term stability of perovskite solar cells. The current strategy primarily employs blocking layers atop the perovskite film to suppress this migration. However, the layers cannot completely prevent ion movement due to trade-offs with carrier transport. In this work, we quantify the barrier energy required to prevent the iodide ions migration from the perovskite film and develop a composite layer that leverages scattering and drift effects on perovskite surface to meet this threshold, reducing ions migration by 99.9%. Moreover, we utilize Poly(N-vinylcarbazole) with a high work function as hole transport material to address the band shift caused by the drift electric-field, thus enhancing hole extraction efficiency. Eventually, the device achieves a certified steady-state efficiency of 25.7% and maintains >95% of the initial efficiency after 1500 hours at 85 °C under maximum power point tracking.

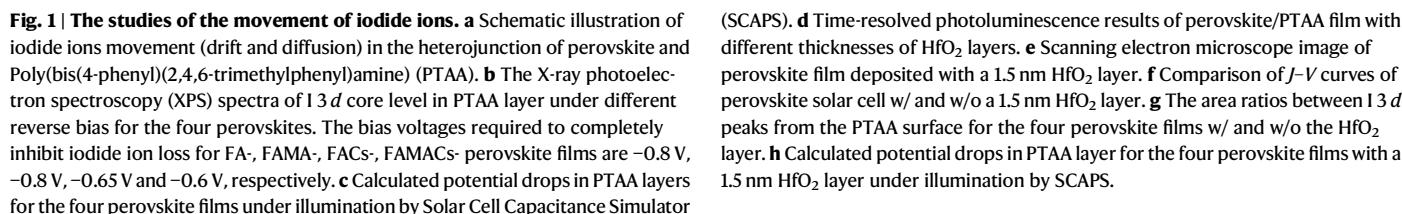
While the power conversion efficiency (PCE) of perovskite solar cells (PSCs) has significantly improved^{1–5}, their operational stability remains insufficient for commercial applications^{6–8}. Ion migration is one of the main causes of device performance degradation during long-term operation since it not only speeds up material deterioration but also triggers interfacial chemical reactions^{9–11}. This process is accelerated by external factors like light exposure and thermal stress, while intrinsic factors like defect-assisted migration pathways and the low activation energy for ion migration within perovskite crystals are the main sources of ion migration^{12–17}.

Iodide ions migrating from the perovskite film into the charge transport layer and electrodes can alter the physical and chemical properties of these materials, therefore disrupting the interface structure and electric-field distribution^{13,18–20}. The current primary strategy to suppress interfacial migration of iodide ions is incorporating a blocking interlayer atop the perovskite film^{21–27}. Various materials with low ion diffusion coefficients have been employed on the perovskite film surface to inhibit ion migration, such as graphene²¹, low-dimensional perovskites^{28,29} and MoS₂³⁰, yet these materials cannot completely suppress iodide ion migration and eliminate the influences^{21,29,30}, eventually leading to decreased stability of perovskite devices. An optimal blocking material requires not only possessing

robust chemical stability but also achieving the balance between inhibiting ion diffusion and maintaining efficient charge carrier transport^{6,22}. The blocking layer with excessive thickness will impede carrier transport, while insufficient thickness lacks the necessary uniformity and blocking capability to effectively prevent ion diffusion. However, quantitative research on suppressing ion interfacial migration is currently lacking, resulting in insufficient theoretical guidance for selecting appropriate blocking materials.

In this work, we determined the threshold barrier energy for suppressing iodide ions migration by balancing its drift and diffusion movement at the perovskite/hole transport layer (HTL) interface. On this basis, we constructed a modulation structure atop the perovskite film, consisting of an atomic-layer-deposited HfO₂ layer and an ordered dipole monolayer. In this structure, the HfO₂ layer blocks the iodide ions through a scattering mechanism and provides anchoring sites for the dipole monolayer^{31,32}, allowing it to form a dense and uniform drift electric-field. This bilayer provides a barrier energy that meets the necessary threshold to restrict interfacial iodide migration. Moreover, we utilized Poly(N-vinylcarbazole) (PVK) with a high work function as the HTL to address the band shift induced by the interfacial electric-field, thereby facilitating the efficient hole extraction. As a result, the PSCs based on FAPbI₃ achieved a PCE of 25.86%, with a

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We utilized Poly(bis(4-phenyl)(2,4,6-trimethylphenyl)amine) (PTAA) as the HTL and first analyzed iodide ions diffusion in aged devices (stored under illumination for 500 h) using time-of-flight secondary ion mass spectrometry (TOF-SIMS), which revealed a significant accumulation of iodide on the PTAA surface (Supplementary Fig. 2). To further characterize iodide migration, we employed X-ray

photoelectron spectroscopy (XPS) to monitor the changes in iodine content at the PTAA interface under reverse bias. The following four perovskite compositions were investigated: FAPbI₃, FA_{0.9}MA_{0.1}PbI₃, FA_{0.9}CS_{0.1}PbI₃, and FA_{0.9}MA_{0.05}CS_{0.05}PbI₃, where FA is Formamidinium and MA is Methylammonium. Taking the FAPbI₃ device as the example, after storing under illumination for 500 h without bias, a strong I 3*d* peaks appeared on the PTAA surface (Fig. 1b). When a reverse bias of −0.8 V was applied, the I 3*d* peaks disappeared from the PTAA surface, indicating that the iodide ions were confined within the PTAA depletion region (Fig. 1b). Moreover, no I 3*d* peak was observed from the PTAA surface when extending the aging time under −0.8 V from 500 to 2000 h (Supplementary Fig. 3), suggesting that the diffusion and drift of iodide ions at the interface reached the equilibrium under this bias condition.

By calculating the built-in field potential under illumination, we found a potential drop of 0.911 V within the PTAA in the depletion region of the perovskite/PTAA interface when −0.8 V bias was applied (Fig. 1c, Tables S1 and S2). This suggests that 0.911 eV of barrier energy is needed for PSCs based on FAPbI₃ to prevent the loss of iodide ions from the perovskite. We applied the same method to test the other perovskite devices (Fig. 1b), and their barrier energy results are presented in Fig. 1c and Table S1. The differences in barrier energy among the various perovskite films can likely be attributed to variations in composition and defect density (Supplementary Fig. 6).

However, because perovskite solar cells cannot function under reverse bias, we designed a composite strategy that combines a scattering blocking layer and an ordered dipole monolayer with drift electric-field to replicate this barrier effect atop the perovskite to suppress iodide ion migration.

Scattering barrier preparation

First, we deposited a layer of HfO₂ on the perovskite surface using the atomic-layer-deposition method; details regarding the fabrication and characterization are provided in Supplementary Note 2. And the investigation of the blocking effects of HfO₂ layer is provided in Supplementary Note 3. As indicated by the time-resolved photoluminescence (TRPL) measurements, the application of a 1.5 nm thick HfO₂ layer did not affect the charge carrier transport between the perovskite and PTAA (Fig. 1d), due to the tunneling effect of carriers through the HfO₂ layer. Moreover, the scanning electron microscope (SEM) images confirmed that the 1.5 nm HfO₂ layer is uniformly deposited on the perovskite film surface (Fig. 1e and Supplementary Fig. 8), and the PCE of the device remained unchanged after the HfO₂ deposition (Fig. 1f).

Subsequent analysis of iodide migration after 500 h of light illumination revealed a 30%–50% reduction in iodide diffusion across the four perovskite compositions (Fig. 1g and Fig. S13), demonstrating the preliminary blocking effect of the HfO₂ layer via ion scattering. Furthermore, we identified the required bias values to suppress ion loss by applying reverse bias to the PSCs based on the four films, as shown in Supplementary Fig. 14. We then calculated the potential drop within the PTAA in the depletion region of the perovskite/PTAA interface under the corresponding reverse bias (Fig. 1h and Table S1). This potential drop represents the energy threshold required to inhibit iodide ion migration from the perovskite film after depositing a 1.5 nm HfO₂ blocking layer.

Drift barrier preparation

As shown in Fig. 1h, the threshold energies required to suppress ion migration in all device types are less than 0.6 eV. This suggests a barrier larger than 0.6 eV must be built on the perovskite surface. Dipole molecules can generate a directional electric-field to provide this suppressive barrier. Our previous research has demonstrated that the dense covalently bonded hydroxyl groups on atomic-layer-deposited

metal oxide surfaces can stably and densely anchor self-assembled molecules⁸. Therefore, we prepared an ordered self-assembled dipole monolayer on the surface of the HfO₂ layer to create a dense and uniform interfacial electric-field on the perovskite surface, leveraging the drift effect to block the remaining diffusing iodide ions.

We selected the molecule (4-(2-(trifluoromethyl)pyrimidin-5-yl)phenyl)boronic acid (CF₃-PBAPy) to provide the drift barrier. Detailed molecular information can be found in Supplementary Note 4. The electrostatic potential distribution of the CF₃-PBAPy molecule indicates that the fluorine and nitrogen atoms at the molecular terminal have higher electron cloud density, while the electron density is relatively lower in the anchoring group (Fig. 2a). Therefore, a directional electric-field is established from the anchoring group toward the terminal region, enabling a uniformly oriented CF₃-PBAPy monolayer to inhibit the diffusion of iodide ions.

We investigated the modulation of the perovskite surface potential by the HfO₂/CF₃-PBAPy heterostructure using the Kelvin probe force microscopy (KPFM). The results showed that after self-assembled the CF₃-PBAPy monolayer atop the HfO₂, the work function of the four perovskite thin films increased significantly by 0.60–0.65 eV (Fig. 2b, c). This indicates that the dipole monolayer induces a surface electric-field on the perovskite layer, with a potential drop exceeding 0.6 V that establishes a barrier above the threshold energy.

Notably, the modified perovskite surface not only exhibited a significant increase in work function but also maintained a highly stable root mean square roughness (R_q) in its surface potential distribution (Fig. 2b). This demonstrated the highly conformal and uniform surface electric-field was built on the perovskite surface. This property can be attributed to the dense anchoring sites provided by the atomic-layer-deposited HfO₂, which facilitate the order arrangement of the CF₃-PBAPy molecules.

Photovoltaic performance

The surface electric-field generated by the CF₃-PBAPy molecular dipole layer causes a 0.60–0.65 eV upward shift of the vacuum energy level, resulting in surface potential drop in the HTL layer (Fig. 3a). This potential drop leads to an energy-level mismatch between perovskite and PTAA layer, which severely limits the PCE (Supplementary Fig. 18). To address this issue, we proposed using a HTL with a deeper highest occupied molecular orbital (HOMO) level to realign the energy levels with the perovskite layer.

Poly(vinylcarbazole) (PVK) is a widely used HTL with a deep HOMO level of 5.85 eV and a bandgap of 3.41 eV (Supplementary Figs. 19 and 20). Upon tris(2-(1H-pyrazol-1-yl)-4-tert-butylpyridine)cobalt(III) tris(bis(trifluoromethylsulfonyl)imide) (Co(III)TFSI) doping, the carrier concentration, conductivity, and dielectric constant of PVK are $1.0 \times 10^{17} \text{ cm}^{-3}$, $1.4 \times 10^{-5} \text{ S cm}^{-1}$ (Supplementary Fig. 21), and 3.51 (Supplementary Fig. 22), respectively, aligning these properties with those of PTAA and 2,2',7,7'-tetrakis(N,N-di(4-methoxyphenyl)amino)-9,9'-spirobifluorene (Spiro-OMeTAD)^{33–36}. With the 0.6 eV vacuum energy-level upward shift, the HOMO level of PVK reaches 5.25 eV (Fig. 3a and Supplementary Fig. 19), enabling efficient hole extraction from the CF₃-PBAPy modified perovskite layer as evidenced by the TRPL results (Fig. 3b).

Based on this design, we fabricated a FAPbI₃/HfO₂/CF₃-PBAPy/PVK device (Supplementary Fig. 23), achieving a power conversion efficiency of 25.86% with an open-circuit voltage (*V*_{OC}) of 1.167 V, a short-circuit current density (*J*_{SC}) of 26.10 mA/cm², and a fill factor (FF) of 84.90% (Fig. 3c and Supplementary Fig. 24). The certified steady-state efficiency was 25.70% (Fig. 3C and Supplementary Fig. 25). Moreover, the champion device with an active area of 1 cm² achieved a PCE of 24.50% (*V*_{OC} = 1.162 V, *J*_{SC} = 26.09 mA/cm², FF = 80.83%) (Supplementary Fig. 26). The PSCs based on the other three perovskite films exhibited steady-state efficiency of 25.05%, 25.23%, and 24.72%,

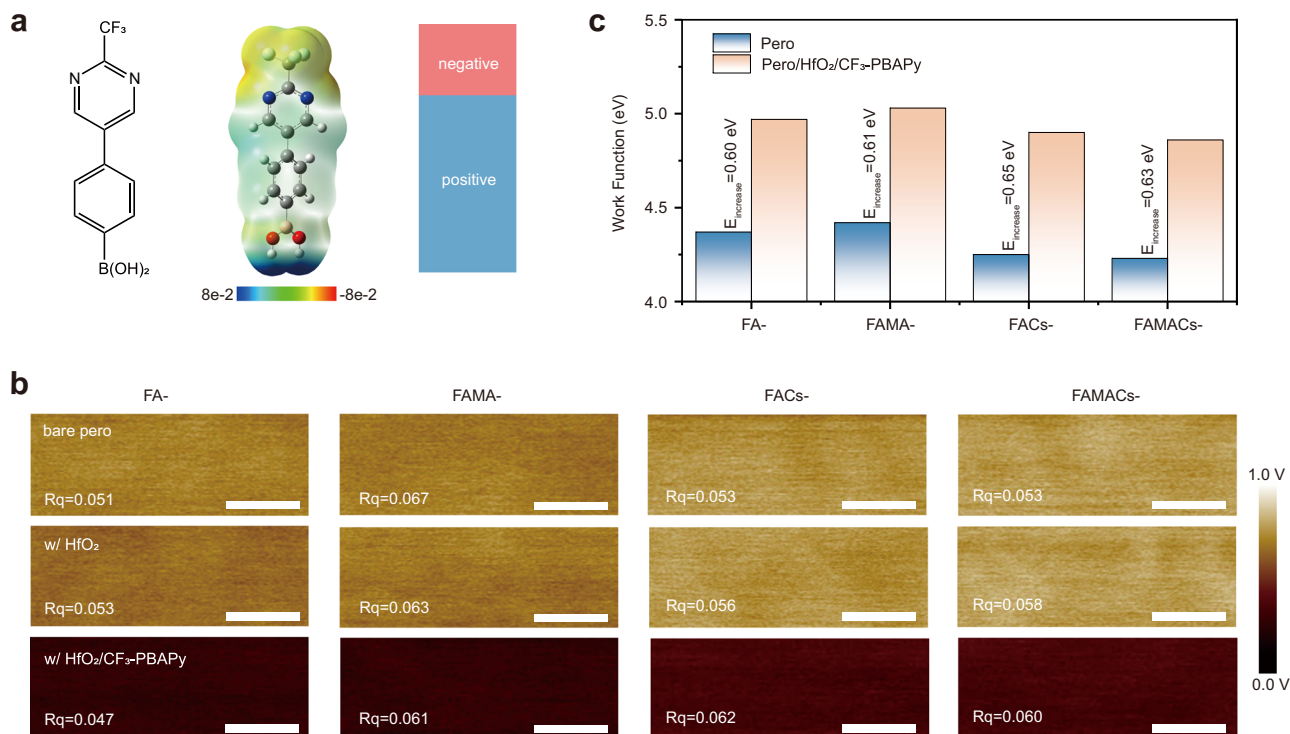


Fig. 2 | Studies on the electric-field in a $\text{CF}_3\text{-PBAPy}$ monolayer. **a Spatial distribution of the electrostatic potential of the $\text{CF}_3\text{-PBAPy}$ molecule. **b** Kelvin probe force microscopy (KPFM) images of perovskite, perovskite/ HfO_2 and perovskite/ $\text{HfO}_2/\text{CF}_3\text{-PBAPy}$ films. The Rq value is the root mean square of the surface potential. Scale bar is 1 μm . **c** Changes of work function after depositing $\text{HfO}_2/\text{CF}_3\text{-PBAPy}$ layer based on the parameters derived from KPFM images.**

respectively, with specific parameters shown in Fig. 3d and Supplementary Figs. 26 and 27.

Inhibition effect for iodide ion migration

To verify the inhibitory effect of the interfacial electric-field on ion migration, we conducted the maximum power point (MPP) degradation tests on all devices under continuous light illumination at 85 °C. We peeled off the HTL from devices that had operated for 1500 h and then used TOF-SIMS to measure the iodine content in the peeled HTL/Au electrode (Fig. 4a, b). In the control sample using PTAA as the HTL, a significant iodine signal was detected. However, for the target sample modified with the $\text{HfO}_2/\text{CF}_3\text{-PBAPy}$ composite layer, the iodine signal intensity in the PVK/Au electrode was only around one-thousandth of that in the control sample. This finding indicates that the composite structure almost completely restricts the outward migration of iodine ions from the perovskite.

Consequently, the target samples exhibited significantly enhanced stability compared to the control devices (Fig. 4c, d, Supplementary Fig. 28). X-ray diffraction (XRD) patterns of the perovskite materials after the stability tests revealed that the diffraction peaks of the target sample experienced negligible attenuation, demonstrating that constraining the iodine outflow during device operation effectively stabilizes the perovskite lattice (Supplementary Fig. 29). Moreover, due to the suppression of iodine ion loss, the devices exhibited exceptional stability in both the humid heat aging test for 1000 h (85 °C/85% relative humidity; ISOS-D-3) (Fig. 4e, f, Supplementary Fig. 30) and the light/dark cycling aging test for 50 cycles (85 °C, 12 h light/12 h dark; ISOS-LC-2) (Fig. 4g, h, Supplementary Fig. 31).

Discussion

We quantified the energy threshold for iodide ion interfacial migration of perovskite films by applying a reverse bias. Through constructing a $\text{HfO}_2/\text{CF}_3\text{-PBAPy}$ composite structure with synergistic scattering and

drift effects atop the perovskite film, a barrier exceeding the energy threshold was achieved, thereby effectively suppressing iodide ion migration and showing significant improvement in operational stability. Moreover, the introduction of PVK with a high work function as the HTL addressed the energy-level offset caused by the drift electric-field, therefore enables highly efficient PSCs.

Methods

Materials

Lead iodide (PbI_2 , 99.99%, trace metal basis), Formamidine hydroiodide (FAI, 99.99%, trace metals basis), Cesium iodide (CsI, >99.0%), Methylamine hydroiodide (MAI, >99.0%), Methylamine hydrochloride (MAH, 98.0%), were purchased from Tokyo Chemical Industry. The solvents including N, N-Dimethylformamide (DMF, 99.8%, anhydrous), Dimethyl sulfoxide (DMSO, 99.9%, anhydrous), Chlorobenzene (CB, 99.8%, anhydrous), Isopropanol (IPA, 99.5%, anhydrous), acetonitrile (ACN, 99.8%, anhydrous) were purchased from Sigma-Aldrich. The tin (II) chloride dihydrate ($\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$, 98%) was purchased from Acros. Urea and deionized water were purchased from Alfa Aesar, potassium chloride (KCl, 99.999%) and thioglycolic acid (TGA, 99%), FK209 Co(III)TFSI salt were purchased from Sigma-Aldrich. The methylenediammonium dichloride (MDACl_2), Propane-1,3-diaminium Iodide (PDAl_2), PVK and PTAA were purchased from Xi'an Yuri Solar Co., Ltd (China).

Perovskite solar cells fabrication

FTO glass substrate were sequentially sonicated in detergent, deionized water, ethanol, acetone, and IPA for 15 min each. The FTO glass were immersed in a solution by mixing $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ (550 mg), urea (2.5 g), TGA (50 μL) and hydrochloric acid with a mass fraction of 37% (2.5 mL) in a 200 mL of deionized water for 8 h at 90 °C. The substrates were then annealed at 150 °C for 1 h, followed by spin-coating with KCl (3 mg/mL in deionized water) and annealing at 150 °C for 20 min. The four perovskite precursor solutions were prepared by dissolving FAI,

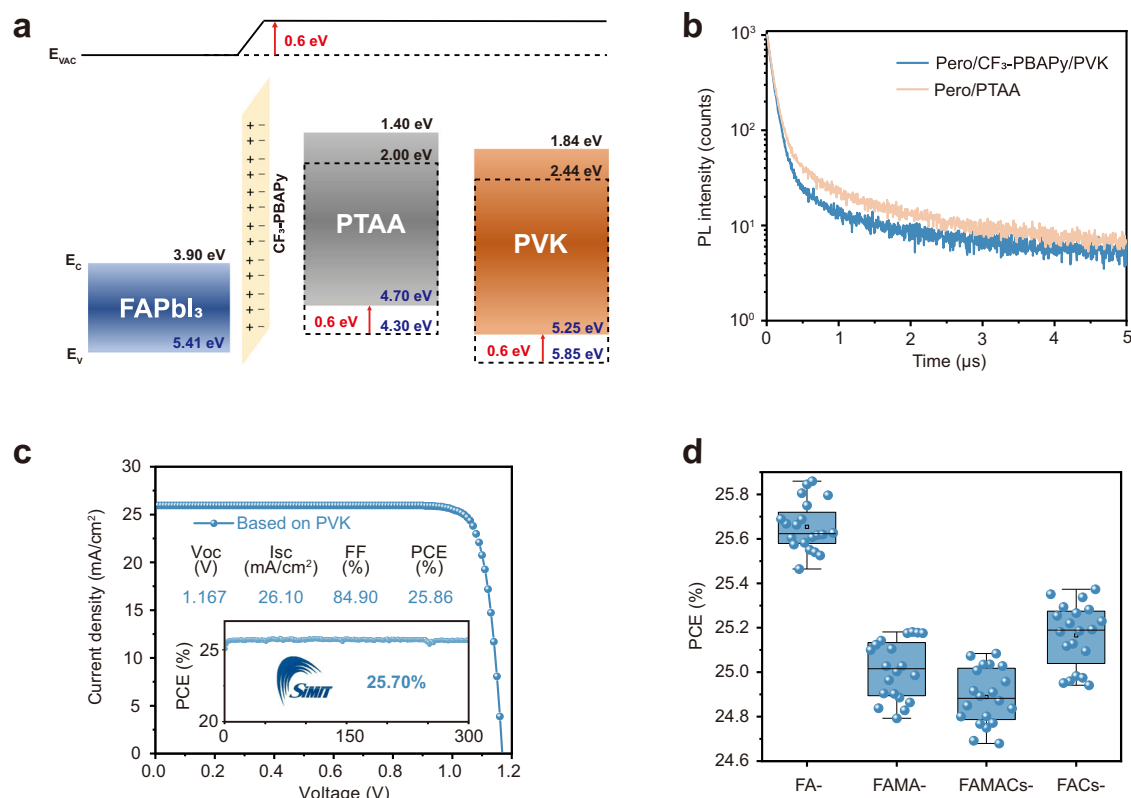


Fig. 3 | Photovoltaics performance. **a** Energy-level diagrams for different hole transport materials (HTMs) and perovskite. **b** Time-resolved photoluminescence results of FAPbI₃ film with different treatment. **c** *J*-*V* curve of PSC based on FAPbI₃ film using Poly(vinylcarbazole) as HTM. The inset shows the certified steady-state

efficiency of the perovskite solar cell (PSC). **d** Power conversion efficiency distribution of PSCs based on different perovskite films (20 devices for each type). The box plot denotes minima (bottom line), maxima (top line), median (center line), and 75th (top edge of the box) and 25th (bottom edge of the box) percentiles.

MAI, CsI and PbI₂ in DMF/DMSO (4:1, 1.5 M) according to their chemical formula. MAI (20 mol%) was added in the solution. For the fabrication of FAPbI₃ film, MDACl₂ (3.0 mol%) was added in the solution to stabilize the α -FAPbI₃ phase. The perovskite solutions were spin-coated onto the substrate at 1000 rpm for 10 s and 5000 rpm for 40 s, and 200 ml of ethyl acetate was dropped at 10 s before the end of the spin-coating. Then the perovskite films were annealed at 120 °C for 20 min, followed by spin-coating PDAI₂ (0.5 mg/mL in IPA/CB = 1:1, 5000 r.p.m.) to passivate the surface defects and prevent the perovskite films from being damaged during the atomic-layer-deposition process. The deposited film was transferred to the ALD system (PicosumTM R-200 Std) to deposit HfO₂ layer. And then CF₃-PBAPy (0.5 mg/mL in IPA, 5000 r.p.m.) was spin-coated, followed by annealing at 100 °C for 5 min. For the HTL preparation, the PTAA and PVK solution (1 mL, 20 mg/mL in CB) was mixed with Co (III) TFSI (40 μ L, 1 mM in ACN), and then the mixed solution was spin-coated (3000 r.p.m.) on the deposited film. The Au electrode (100 nm) was deposited by thermal evaporation.

Stability tests

The PSCs were encapsulated in the nitrogen-filled glove box with O₂ and H₂O level <0.1 ppm. A glass slide was covered over the PSCs with edges sealed by butyl tape, followed by vacuum lamination at 115 °C for 15 min. The maximum power point tracking measurement was performed under AM 1.5 G illumination (100 mW cm⁻²) at 85 °C in ambient air. The damp-heat stability test was performed on encapsulated devices in an aging box with approximately 85 °C and 85% RH. The light/dark cycling stability test was performed under AM 1.5 G illumination (100 mW cm⁻²) at 85 °C in ambient air, the PSCs were exposed in simulated sunlight which turned on and off with cycle periods of 24 h and duty cycles (light: dark) of 1:1.

Relative dielectric constant

The electrochemical impedance spectrum was performed to calculate the relative dielectric constant (ϵ_r). The structure of testing sample is ITO/perovskite/Au or ITO/PVK/Au. This measurement conducted in dark with the frequency range between 1 Hz and 1×10^7 Hz. The capacitance (*C*) is calculated by:

$$C = -1/(2\pi f Z'') \dots \quad (1)$$

where *f* is the frequency and *Z''* is the imaginary part of impedance.

The relative dielectric constant (ϵ_r) is calculated by³⁷:

$$\epsilon_r = LC/(A\epsilon_0) \dots \quad (2)$$

where *L* is the thickness of perovskite films (800 nm) or PVK (150 nm), *C* is the capacitance, *A* is the effective area (0.05 cm²), ϵ_0 is the vacuum permittivity with the value of $8.854187817 \times 10^{-12}$ F/m.

Carrier concentration characterization

The carrier concentration is measured by Hall effect testing using the van der Pauw method. The carrier concentration (*n*, *p*) is calculated using the following formula:

$$R_H = 1/qp(\text{for } p - \text{type}) \dots \quad (3)$$

$$R_H = -1/qn(\text{for } n - \text{type}) \dots \quad (4)$$

Where *R_H* is the Hall coefficient, *q* is the electronic charge. In this work, the carrier concentration of PVK is 1×10^{17} cm⁻³, and the values for PTAA and different perovskite films are provided in Table S2.

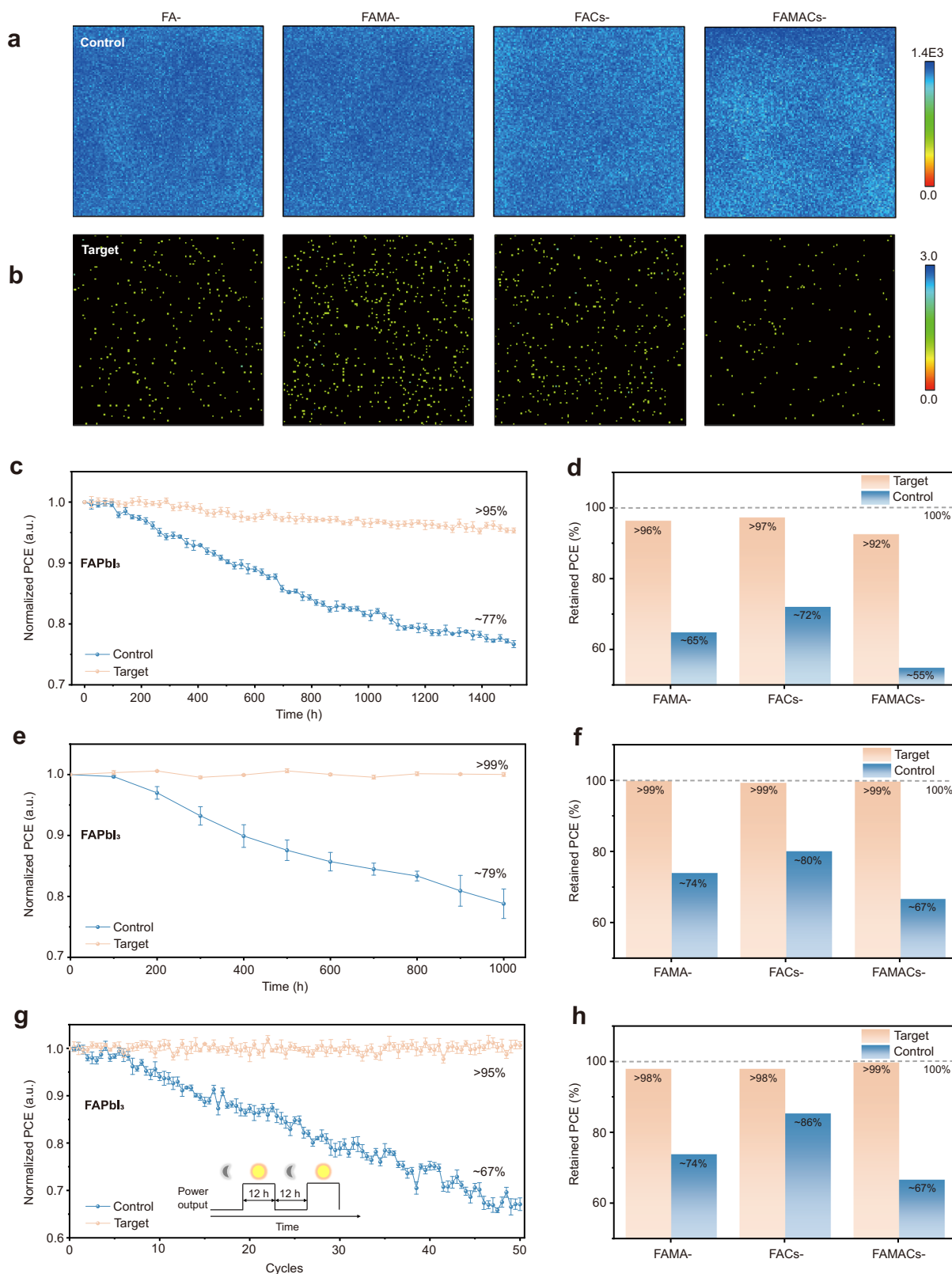


Fig. 4 | Stability performance. **a, b** Time-of-flight secondary ion mass spectrometry mapping of I^- signals in hole transport layer/Au electrode from the aged PSCs based on different perovskite. The analysis area is $150 \times 150 \mu m^2$. **c** Maximum power point (MPP) tracking results of the perovskite solar cells (PSCs) based on FAPbI₃ film, the average initial power conversion efficiency (PCE) of target and control PSCs are 25.58% and 24.11%. **d** MPP tracking results in the PSCs based on perovskite films of FAMAPbI₃, FACSPbI₃, and FAMACSPbI₃. **e** Damp-heat stability results of the

PSCs based on FAPbI₃ film, the average initial PCE of target and control PSCs are 25.68% and 24.19%. **f** MPP tracking results in the PSCs based on perovskite films of FAMAPbI₃, FACSPbI₃, and FAMACSPbI₃. **g** Light/dark cycling stability results of the PSCs based on FAPbI₃ film, the average initial PCE of target and control PSCs are 25.71% and 24.24%. **h** Light/dark cycling stability results in the PSCs based on perovskite films of FAMAPbI₃, FACSPbI₃, and FAMACSPbI₃. The error bars denote standard deviation for four individual devices.

Space charge limited current (SCLC) characterization

The SCLC tests of the perovskite films were performed with electron-only device (FTO/SnO₂/perovskite/PCBM/Au) to obtain the defect density (N_{trap}). The J - V curves of different sample were provided in Supplementary Fig. 6. At a low bias voltage, the linear J - V curve is the ohmic region. At an intermediate bias voltage, a trap-filled region is reached at a voltage (V_{TFL}) where the current shows a rapid nonlinear rise and transfers into a trap-filled limit (TFL). In this region, the N_{trap} can be expressed as³⁸:

$$N_{\text{trap}} = 2V_{\text{TFL}}\epsilon\epsilon_0/(eL^2) \dots \quad (5)$$

where V_{TFL} is the TFL voltage, ϵ is the relative dielectric constant of perovskite, ϵ_0 is the vacuum permittivity, e is the electronic charge, and L is the thickness of the perovskite (800 nm).

The SCLC tests of the PVK films were performed with hole-only device (ITO/PVK/Au) to obtain the hole mobility³⁹. The J - V^2 curve of PVK sample was provided in Supplementary Fig. 21. At the high bias voltage, the curve is the child region. The carrier mobility can be calculated by fitting the child region using Mott-Gurney law:

$$J = 9\epsilon_0\epsilon_r\mu V^2/(8L^3) \dots \quad (6)$$

where J is the current density, ϵ_0 is the vacuum permittivity, ϵ_r is the relative dielectric constant, μ is the carrier mobility, V is the voltage, L is the thickness of PVK films (150 nm).

SCAPS simulation

Simulation of our PSCs were carried out using SCAPS. The parameters used in the simulation are provided in Table S2, in which the energy band information was obtained from Supplementary Figs. 4 and 5 measured by UPS tests and UV-vis absorption spectroscopy. And the shallow uniform donor density (N_D) and acceptor density (N_A) were measured by Hall tests. The relative dielectric constants of different materials were obtained according to the description above. The rest of the parameters are derived from the reference.

Fitting of Fick's second law of diffusion

The diffusion of iodide ions in HfO₂ layer follows the Fick's second law, where the perovskite side can be regarded as a diffusion medium without composition change. In the initial state, the iodide ion concentration in the perovskite layer is ρ_0 , while it is 0 in HfO₂. After 500 h of diffusion, the iodide ion concentration in HfO₂ at 1.5 nm obeys:

$$\rho(x, t) = \rho_0 - \rho_0 \times \text{erf}(x/2\sqrt{Dt}) \dots \quad (7)$$

Where the erf is the Gauss error function. We fitted the diffusion of iodide ions according to the relation above, and the results were presented in Supplementary Fig. 10.

J - V characterization

J - V curves of the devices were recorded by a Keithley 2400 digital source measurement and a solar simulator (J - V Newport, Oriel Class A, 91195A) with an AM 1.5G spectrum. The light intensity was calibrated by a silicon-reference solar cell certified by Research Center for Photovoltaics (Advanced Industrial Science and Technology, AIST, Japan). The J - V tests were performed forward scan (from -0.2 to 1.2 V) or reverse scan (from 1.2 to -0.2 V), with the delay time and the step voltage of 200 ms and 10 mV respectively.

Characterization

NMR spectra were collected by AVANCE III (Bruker Co., Germany, frequency of 400 MHz). The X-ray diffraction patterns were measured by D8 ADVANCE Da Vinci (Bruker Co., Germany) with the Cu K α

radiation ($\lambda = 1.54056 \text{ \AA}$). UV-vis absorption spectra of perovskite films were measured by a Shimadzu UV 2450 spectrometry. The PL and TRPL spectrums were performed on steady-state transient fluorescence spectrometer (FLS1000). The XPS was conducted with a Kratos AXIS Ultra DLD by Al K α X-ray source. UPS was conducted with a Kratos AXIS Ultra DLD using He I α ($h\nu = 21.22 \text{ eV}$). The time-of-flight secondary ion mass spectroscopy (TOF-SIMS) was performed on TOF-SIMS 5-100 (ION-TOF GmbH). Bi³⁺ (30 keV) was used to analyze deeply about the sample with an analysis area of $150 \times 150 \mu\text{m}^2$. KPFM tests were conducted by atomic force microscope (MFP-3D). Hall tests were performed by PPMS-9T(EC-II) (Quantum Design).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings are available in the Article and its Supplementary Information. Source data are provided with this paper.

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Author contributions

H.T. and Q.H. conceived the idea. H.T. and Y.S. conducted the theoretical calculations. H.T., G.Y. and Y.S. performed the characterizations of films and devices and analyzed the corresponding results. H.T. and Q.H. wrote the paper. L.H. and Q.H. reviewed and edited the manuscript. All authors discussed the results provided in this paper.

Competing interests

The authors declare no competing interests.

Additional information

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